MATH329 · OPERATIONS RESEARCH

# Lecture 5 Big-M Method

Midterm summary · Chapter 3 · Artificial variable techniques

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## 01 Why do we need Big-M?

**Simplex** starts from the slack variables, because they give the identity matrix directly. That happens when every constraint is `≤`. A `≥` constraint gets a surplus with a minus sign, and an `=` constraint gets nothing, so there is no identity matrix to start from.

> LPP in which all the constraints are (≤) with nonnegative right-hand sides offer a convenient all-slack starting basic feasible solution. Models involving (=) and/or (≥) constraints do not.
> The procedure for starting "ill-behaved" LPP with (=) and (≥) constraints is to use artificial variables that play the role of slacks at the first iteration, and then dispose of them legitimately at a later iteration.

The fix: add a temporary variable called an **Artificial variable**, written `R`, to every `≥` or `=` constraint. It completes the identity matrix, so we can start.

| Constraint | What we add |
|---|---|
| `≤` | A slack `+s` only |
| `≥` | A surplus `−s` and an artificial `+R` |
| `=` | An artificial `+R` only |

## 02 The penalty M

`R` is not part of the original problem, so it must leave and equal zero in the final solution. To force it out, we give it a huge coefficient `M` in the objective (a very large positive

number):

The artificial variables are assigned a very high penalty in the objective function, thus forcing them (eventually) to equal zero in the optimum solution:
= –M in a maximization problem
= M in a minimization problem
where M is a very large positive number.

| Problem | Coefficient of R in the objective | Why |
|---|---|---|
| Max | `−M` | Any positive `R` lowers `z` a lot, so Max drives it out. |
| Min | `+M` | Any positive `R` raises `z` a lot, so Min drives it out. |

## 03 The steps

| # | Step |
|---|---|
| 1 | Write the standard form: a slack for `≤` and a surplus for `≥`. |
| 2 | Add `R` to every `≥` or `=` constraint, and add `−MR` (max) or `+MR` (min) to the objective. |
| 3 | Write the first tableau. The `Z` row comes from `z − (objective) = 0`, so every coefficient flips sign: `R` has `M` under it in max and `−M` in min. |
| 4 | Fix the `Z` row: `R` is basic, so its coefficient in the `Z` row must become zero (a requirement to start simplex). Use the formula below. |
| 5 | Continue simplex as usual with the rules of Lecture 4. |

( New z-row = Old z-row + (M·R1-row + M·R2-row) )  min

( New z-row = Old z-row − (M·R1-row + M·R2-row) )  max

**How to compare values with M**

`M` is bigger than any number, so compare the coefficient of `M` first. For example `−2 + 4M` is larger than `−1 + 2M` because `4M > 2M`. If the `M` parts are equal, compare the plain number. For min the most positive enters, and for max the most negative enters, as in Lecture 4.

## 04 Example 1: min

| |
|---|
| Minimize $z = 2x_1 + x_2$ |
| subject to $3x_1 + x_2 \geq 9$ |
| $x_1 + x_2 \geq 6$ |
| $x_1, x_2 \geq 0$ |

### 1 Standard form + artificial

| |
|---|
| Minimize $z = 2x_1 + x_2 + 0s_1 + 0s_2 + MR_1 + MR_2$ |
| subject to $3x_1 + x_2 - s_1 + R_1 = 9$ |
| $x_1 + x_2 - s_2 + R_2 = 6$ |
| $x_1, x_2, s_1, s_2, R_1, R_2 \geq 0$ |

## 2 The tableau before the fix

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | $R_1$ | $R_2$ | Sol. |
|---|---|---|---|---|---|---|---|
| $R_1$ | 3 | 1 | −1 | 0 | 1 | 0 | 9 |
| $R_2$ | 1 | 1 | 0 | −1 | 0 | 1 | 6 |
| Z | −2 | −1 | 0 | 0 | −M | −M | 0 |

$R_1$ and $R_2$ have $-M$ under them, and they must be zero. It is a min problem, so we add:

```
New Z = Old Z + M(R₁ row) + M(R₂ row)
```

```
x₁:  −2 + M(3) + M(1) = −2 + 4M      Sol.:  0 + 9M + 6M = 15M
```

## 3 The iterations

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | $R_1$ | $R_2$ | Sol. |
|---|---|---|---|---|---|---|---|
| $R_1$ | 3 | 1 | $-1$ | 0 | 1 | 0 | 9 |
| $R_2$ | 1 | 1 | 0 | $-1$ | 0 | 1 | 6 |
| Z | $4M - 2$ | $2M - 1$ | $-M$ | $-M$ | 0 | 0 | $15M$ |
| $x_1$ | 1 | $\frac{1}{3}$ | $-\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 0 | 3 |
| $R_2$ | 0 | $\frac{2}{3}$ | $\frac{1}{3}$ | $-1$ | $-\frac{1}{3}$ | 1 | 3 |
| Z | 0 | $\frac{2M}{3} - \frac{1}{3}$ | $\frac{M}{3} - \frac{2}{3}$ | $-M$ | $\frac{2}{3} - \frac{4M}{3}$ | 0 | $3M + 6$ |
| $x_1$ | 1 | 0 | $-\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $-\frac{1}{2}$ | $\frac{3}{2}$ |
| $x_2$ | 0 | 1 | $\frac{1}{2}$ | $-\frac{3}{2}$ | $-\frac{1}{2}$ | $\frac{3}{2}$ | $\frac{9}{2}$ |
| Z | 0 | 0 | $-\frac{1}{2}$ | $-\frac{1}{2}$ | $\frac{1}{2} - M$ | $\frac{1}{2} - M$ | $\frac{15}{2}$ |

- Iteration 1: the most positive is $-2 + 4M$ under $x_1$. The ratios are $\frac{9}{3} = 3$ and $\frac{6}{1} = 6$, so $R_1$ leaves.
- Iteration 2: the most positive is under $x_2$. The ratios are $\frac{3}{\frac{1}{3}} = 9$ and $\frac{3}{\frac{2}{3}} = 4.5$, so $R_2$ leaves.
- Every entry in the $Z$ row is now $\leq 0$ (since $M$ is large), so we stop.

$$\text{Min } z = \frac{15}{2} \quad \text{at} \quad x_1 = \frac{3}{2}, \quad x_2 = \frac{9}{2}$$

$$\text{Check:} \quad 2\left(\frac{3}{2}\right) + \frac{9}{2} = 3 + 4.5 = 7.5 \ \checkmark$$

## 05 Example 2: max with an = constraint

$$\begin{aligned} &\text{Maximize } z = 2x_1 + 3x_2 - 5x_3 \\ &\text{subject to} \quad x_1 + x_2 + x_3 = 7 \\ &\qquad\qquad 2x_1 - 5x_2 + x_3 \geq 10 \\ &\qquad\qquad x_1, x_2, x_3 \geq 0 \end{aligned}$$

### 1 Standard form + artificial

The first constraint is $=$, so it takes only $R_1$. The second is $\geq$, so it takes $-s_2$ and $R_2$.

$$\begin{aligned} &\text{Maximize } z = 2x_1 + 3x_2 - 5x_3 + 0s_2 - MR_1 - MR_2 \\ &\text{subject to} \quad x_1 + x_2 + x_3 + R_1 = 7 \\ &\qquad\qquad 2x_1 - 5x_2 + x_3 - s_2 + R_2 = 10 \end{aligned}$$

### 2 The tableau before the fix

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_2$ | $R_1$ | $R_2$ | Sol. |
|---|---|---|---|---|---|---|---|
| $R_1$ | 1 | 1 | 1 | 0 | 1 | 0 | 7 |
| $R_2$ | 2 | −5 | 1 | −1 | 0 | 1 | 10 |
| Z | −2 | −3 | 5 | 0 | M | M | 0 |

It is a max problem, so we subtract: `New Z = Old Z − M(R₁ row) − M(R₂ row)`.

## 3 The iterations

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_2$ | $R_1$ | $R_2$ | Sol. |
|---|---|---|---|---|---|---|---|
| $R_1$ | 1 | 1 | 1 | 0 | 1 | 0 | 7 |
| $R_2$ | **2** | −5 | 1 | −1 | 0 | 1 | 10 |
| $Z$ | $-3M-2$ | $4M-3$ | $5-2M$ | $M$ | 0 | 0 | $-17M$ |
| $R_1$ | 0 | $\frac{7}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 1 | $-\frac{1}{2}$ | 2 |
| $x_1$ | 1 | $-\frac{5}{2}$ | $\frac{1}{2}$ | $-\frac{1}{2}$ | 0 | $\frac{1}{2}$ | 5 |
| $Z$ | 0 | $-\frac{7M}{2}-8$ | $6-\frac{M}{2}$ | $-\frac{M}{2}-1$ | 0 | $\frac{3M}{2}+1$ | $10-2M$ |
| $x_2$ | 0 | 1 | $\frac{1}{7}$ | $\frac{1}{7}$ | $\frac{2}{7}$ | $-\frac{1}{7}$ | $\frac{4}{7}$ |
| $x_1$ | 1 | 0 | $\frac{6}{7}$ | $-\frac{1}{7}$ | $\frac{5}{7}$ | $\frac{1}{7}$ | $\frac{45}{7}$ |
| $Z$ | 0 | 0 | $\frac{50}{7}$ | $\frac{1}{7}$ | $M+\frac{16}{7}$ | $M-\frac{1}{7}$ | $\frac{102}{7}$ |

- Iteration 1: the most negative is $-2-3M$ under $x_1$. The ratios are $\frac{7}{1}=7$ and $\frac{10}{2}=5$, so $R_2$ leaves.
- Iteration 2: the most negative is under $x_2$. The $x_1$ row has a negative coefficient, so we skip it, and $R_1$ leaves.

$$\text{Max } z=\frac{102}{7} \quad \text{at} \quad x_1=\frac{45}{7},\ x_2=\frac{4}{7},\ x_3=0$$

## 06 Example 3: min with three constraints

$$\begin{aligned} &\text{Minimize } z = 4x_1 + 6x_2 \\ &\text{subject to} \quad -2x_1 + 3x_2 = 3 \\ &\qquad\qquad 4x_1 + 5x_2 \geq 10 \\ &\qquad\qquad 4x_1 + 8x_2 \geq 5 \\ &\qquad\qquad x_1, x_2 \geq 0 \end{aligned}$$

$$\begin{aligned} &\text{Minimize } z = 4x_1 + 6x_2 + MR_1 + MR_2 + MR_3 \\ &\text{subject to} \quad -2x_1 + 3x_2 + R_1 = 3 \\ &\qquad\qquad 4x_1 + 5x_2 - s_2 + R_2 = 10 \\ &\qquad\qquad 4x_1 + 8x_2 - s_3 + R_3 = 5 \end{aligned}$$

| Basic | $x_1$ | $x_2$ | $s_2$ | $s_3$ | $R_1$ | $R_2$ | $R_3$ | Sol. |
|---|---|---|---|---|---|---|---|---|
| $R_1$ | $-2$ | $3$ | $0$ | $0$ | $1$ | $0$ | $0$ | $3$ |
| $R_2$ | $4$ | $5$ | $-1$ | $0$ | $0$ | $1$ | $0$ | $10$ |
| $R_3$ | $4$ | **$8$** | $0$ | $-1$ | $0$ | $0$ | $1$ | $5$ |
| $Z$ | $6M - 4$ | $16M - 6$ | $-M$ | $-M$ | $0$ | $0$ | $0$ | $18M$ |
| $R_1$ | $-\frac{7}{2}$ | $0$ | $0$ | **$\frac{3}{8}$** | $1$ | $0$ | $-\frac{3}{8}$ | $\frac{9}{8}$ |
| $R_2$ | $\frac{3}{2}$ | $0$ | $-1$ | $\frac{5}{8}$ | $0$ | $1$ | $-\frac{5}{8}$ | $\frac{55}{8}$ |
| $x_2$ | $\frac{1}{2}$ | $1$ | $0$ | $-\frac{1}{8}$ | $0$ | $0$ | $\frac{1}{8}$ | $\frac{5}{8}$ |
| $Z$ | $-2M - 1$ | $0$ | $-M$ | $M - \frac{3}{4}$ | $0$ | $0$ | $\frac{3}{4} - 2M$ | $8M + \frac{15}{4}$ |
| $s_3$ | $-\frac{28}{3}$ | $0$ | $0$ | $1$ | $\frac{8}{3}$ | $0$ | $-1$ | $3$ |
| $R_2$ | **$\frac{22}{3}$** | $0$ | $-1$ | $0$ | $-\frac{5}{3}$ | $1$ | $0$ | $5$ |
| $x_2$ | $-\frac{2}{3}$ | $1$ | $0$ | $0$ | $\frac{1}{3}$ | $0$ | $0$ | $1$ |
| $Z$ | $\frac{22M}{3} - 8$ | $0$ | $-M$ | $0$ | $2 - \frac{8M}{3}$ | $0$ | $-M$ | $5M + 6$ |
| $s_3$ | $0$ | $0$ | $-\frac{14}{11}$ | $1$ | $\frac{6}{11}$ | $\frac{14}{11}$ | $-1$ | $\frac{103}{11}$ |
| $x_1$ | $1$ | $0$ | $-\frac{3}{22}$ | $0$ | $-\frac{5}{22}$ | $\frac{3}{22}$ | $0$ | $\frac{15}{22}$ |
| $x_2$ | $0$ | $1$ | $-\frac{1}{11}$ | $0$ | $\frac{2}{11}$ | $\frac{1}{11}$ | $0$ | $\frac{16}{11}$ |
| $Z$ | $0$ | $0$ | $-\frac{12}{11}$ | $0$ | $\frac{2}{11} - M$ | $\frac{12}{11} - M$ | $-M$ | $\frac{126}{11}$ |

- Iteration 1: $x_2$ enters ($-6 + 16M$). The ratios are $1$, $2$ and $\frac{5}{8}$, so $R_3$ leaves.
- Iteration 2: the only positive entry is $M - \frac{3}{4}$ under $s_3$, so the surplus enters. The ratios are $3$ and $11$, so $R_1$ leaves.

- Iteration 3: $x_1$ enters. The only row with a positive coefficient is $R_2$, so it leaves.

$$\text{Min } z = \frac{126}{11} \quad \text{at} \quad x_1 = \frac{15}{22}, \quad x_2 = \frac{16}{11}$$

$$\text{Check: } -2\left(\frac{15}{22}\right) + 3\left(\frac{16}{11}\right) = -\frac{15}{11} + \frac{48}{11} = 3 \checkmark$$

Note on the slides (checked by calculation): in Example 1 the last tableau has $\frac{1}{2} - M$ under $R_1$ and $R_2$ (the slide shows $\frac{1}{2} - \frac{M}{2}$). In Example 3: the second tableau has $\text{Sol.} = \frac{15}{4} + 8M$ (the slide shows $\frac{5}{4} + 8M$), the third tableau has $2 - \frac{8M}{3}$ under $R_1$ (the slide shows $2 + \frac{8M}{3}$), and the last tableau has $-\frac{12}{11}$ under $s_2$ (the slide shows $-\frac{6}{11}$) and $x_1 = \frac{15}{22}$ (the table shows $\frac{15}{11}$; the text below it is correct). The final answer does not change.

## 07 Remarks from the slides

1. If in any iteration, there is a tie for **entering** variable between an artificial variable and other variable (decision, surplus or slack), we must prefer the **non-artificial** variable to enter the basis.
2. If in any iteration, there is a tie for **leaving** variable between an artificial variable and other variable, we must prefer the **artificial** variable to leave the basis.
3. If in the final optimal tableau, an artificial variable is present in the basis at a non-zero level, this means our original problem has **no feasible solution**.

| # | In short |
|---|---|
| 1 | A tie for entering: prefer the ordinary variable, not $R$. |
| 2 | A tie for leaving: prefer $R$ to leave. |
| 3 | If $R$ stays in the final tableau with a nonzero value, the problem is **infeasible** (it has no solution). |

## 08 Common mistakes

| Mistake | Correct |
|---|---|
| Using `+M` in max | Max takes `−MR`, min takes `+MR`. |
| Adding `R` to a `≤` constraint | `R` goes only with `≥` and `=`. |
| Forgetting the surplus with `≥` | `≥` takes both `−s` and `+R`. |
| Starting simplex without fixing the `Z` row | The coefficient of `R` in the `Z` row must become zero first. |
| Treating `M` as an ordinary number | The coefficient of `M` decides the comparison. |